

Chapter 168

Resolve FX Transform

The plugins in this category allow different types of animated and non-animated transforms than are possible in the Sizing palette or Clip Transform controls of the Inspector.

Contents

Match Move (Studio Version Only)	3615	Global Blend.....	3640
A Match Move Workflow.....	3615	Video Collage	3640
Main Controls.....	3620	Create Background.....	3640
Tracking Controls.....	3620	Create Tile.....	3641
Positioning.....	3621	Animating Tiles and Advanced Layouts.....	3646
Compositing.....	3622	Video Collage Controls.....	3647
Stabilizing.....	3622	Layout.....	3648
Surface Tracker (Studio Version Only)	3623	Margins and Spacing.....	3648
Bounds.....	3624	Synchronize Keyframing.....	3648
Mesh.....	3624	Manage Tiles.....	3649
Track.....	3625	Mute Tiles.....	3649
Result.....	3627	Custom Size/Shape.....	3649
Using the Surface Tracker.....	3629	Resize Content.....	3650
Transform	3637	Tile Styling.....	3650
General.....	3637	Drop Shadow.....	3650
Position Controls.....	3639	Tile Animation.....	3651
Image Adjustment.....	3639	Easing & Blur.....	3652
Animation.....	3639	Global Blend.....	3652
Advanced Options.....	3639		

Match Move (Studio Version Only)

TIP: Match move workflows are now much more effectively accomplished in the Fusion page using the tools and methods found there.

The Match Move filter is a patch-based tracker that follows a pattern region defined by a tracker control. You can place as many trackers as you like by applying this filter to a node and clicking on a feature you want to track in the Viewer. Ideal features for tracking are high contrast details that have sharp angles that are clearly defined.

Each tracker control consists of (a) a center point that indicates where the tracked motion path will be centered, (b) an inner box that identifies the patch of image you're tracking (that can be resized), (c) an outer box that defines the search region for the track (that can also be resized), and (d) a patch window that shows you a zoomed in closeup of the patch you're tracking. Patch windows can be resized, and the other tracking controls hidden or shown, using controls in the Display Options section of the Match Move controls.



A tracker control in the Match Move filter

If you've accidentally placed a tracker control somewhere you don't want one, you can either drag anywhere within the inner box to move it to another location, or you can Option-click it to delete that tracker.

Once placed, you can choose how much image detail is considered for analysis by resizing that tracker control. To resize a tracker, move the pointer over the edge of the inner box, and drag to shrink or enlarge it.

This plugin will automatically derive as much motion data as the number of trackers you've placed allows:

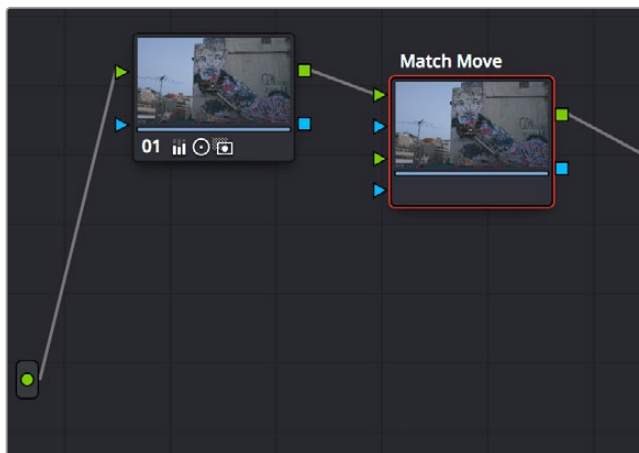
- One tracker tracks horizontal and vertical position (pan and tilt).
- Two or more trackers also track rotation.
- Four or more trackers also track perspective.
- If you place six or more trackers, then the Match Move filter is capable of automatically choosing the best tracker data from the overall collection to give you the most accurate track that's possible.

A Match Move Workflow

The following example shows the simplest complete workflow for importing a foreground clip to composite with a match move to follow a background clip that's already in the timeline. This workflow is suitable for such operations as placing a graphic as a sign on a side of a building, or into the display of a phone.

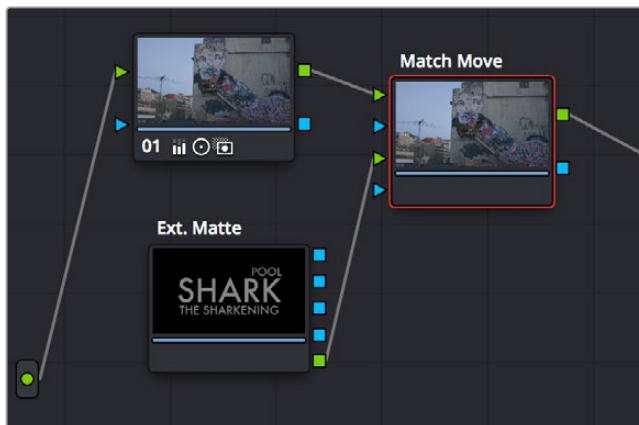
To match move:

- 1 Open the Media Pool in the Edit page, right-click a clip you want to match move to a clip in the Timeline, and choose “Add as Matte For Color Page Clip” to add it to the Color page Node Editor of the currently selected clip as a Timeline Matte. This is currently the only way you can match move and composite a foreground clip against a background clip in the Color page.
- 2 Open the Color page, make sure the background clip you want to match move to is selected, then open the OpenFX library and drag the Match Move plugin directly onto the connection from the last node in the tree to the Output, to add it as a stand-alone FX node.



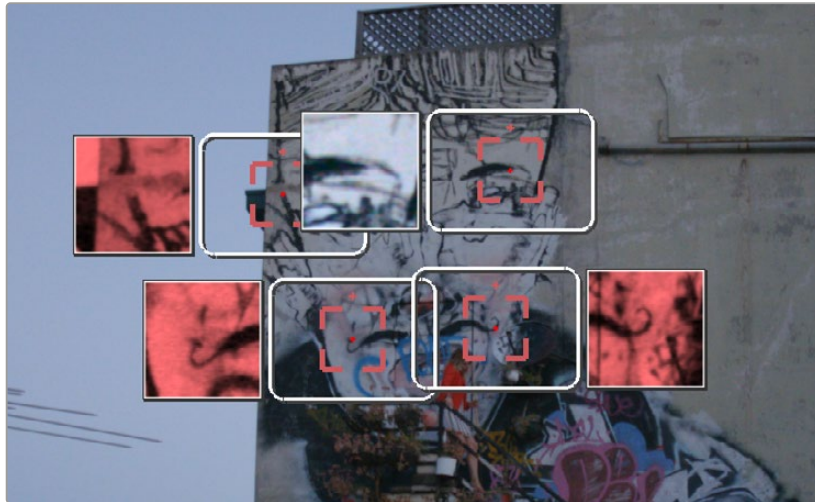
Adding the Match Move filter as a stand-alone FX node

- 3 Right-click any corrector node in the node tree (Node 01 in this example) and choose the clip you added in step 1 in the Add Matte > Timeline Mattes submenu of the contextual menu to expose that clip as an Ext. Matte node. Delete the connection between the new Ext. Matte node and the node it's connected to, and then reconnect its RGB output to the second input of the FX node.



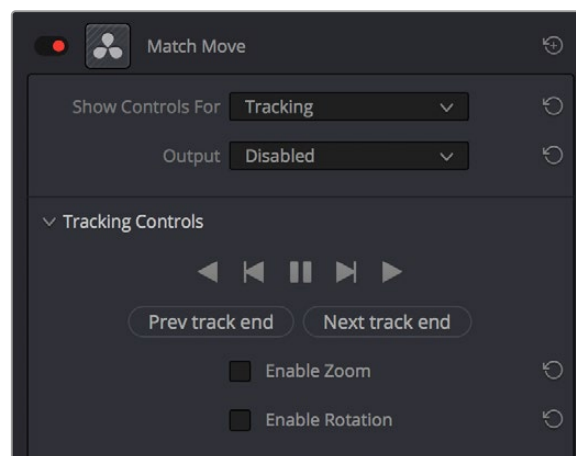
Connecting the foreground image to the second input of the FX node

- 4 Select the FX node, make sure the OpenFX onscreen controls are enabled, choose whether you want to start at the first or last frames of the clip to begin the tracking process, and then click within the Viewer to place tracker controls on the surface of the image you're going to attach the foreground image to. If you want to do a perspective track, place at least four tracker patches in areas you know won't be occluded by other things in the frame. Choose features that are high contrast and angular for the best results, and try to make sure the features you track won't go offscreen or be occluded by other features within the frame. As you place tracking controls, their patch window will be tinted the color of the channel that DaVinci Resolve automatically determines will provide the best tracking result.



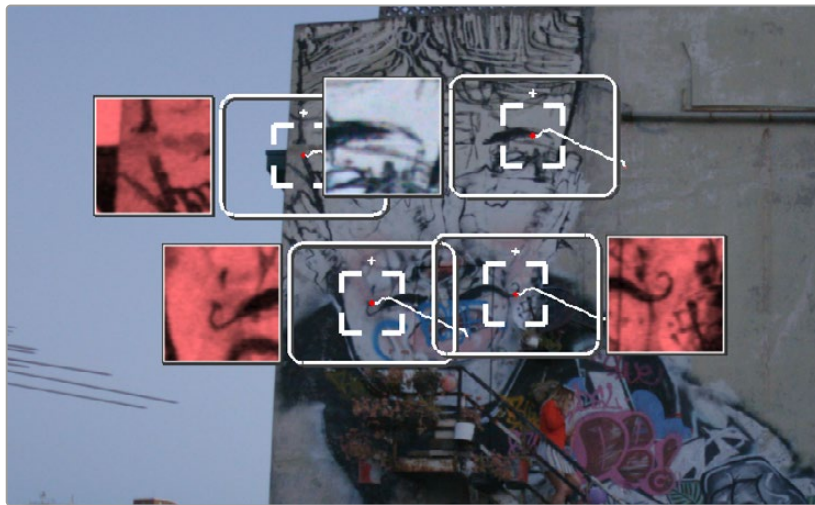
Adding tracker controls to the feature you need to Match Move to

- 5 (Optional) If necessary, you can delete trackers you don't want by Option-clicking them.
- 6 Now that your tracker controls are all set up, you're ready to begin the tracking analysis. If the playhead is at the beginning of the clip, click Track Forward, otherwise click Track Backward. DaVinci Resolve will automatically analyze all tracker controls within the clip until the last frame of the clip is reached.



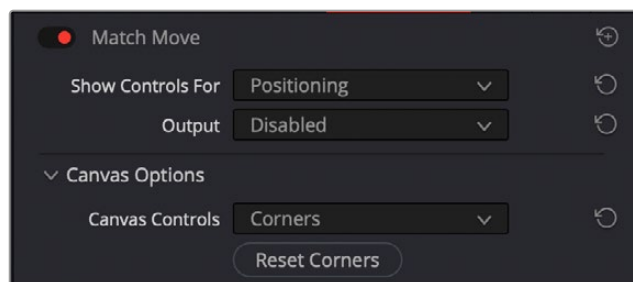
Using the Tracking controls

- 7 (Optional) If anything occludes a patch that's being tracked, or if the tracked feature starts going off the edge of the frame, that tracker control may veer off course and will eventually turn off. If this happens, there's a way to continue the track that's surprisingly easy:
 - a) Click the Track Forward or Track Backward button again to stop the analysis.
 - b) Move the playhead back to the last good frame of tracking data, then click the inner box of each tracker control that went wrong to make sure they're turned on (you know it's on if the patch window is displayed), and then click "Clear tracking forward" or "Clear tracking backward" as necessary to delete the incorrect tracking data. When you're done, click the inner box of each tracker that went wrong to turn it back off again, so it doesn't continue trying to track incorrectly (you know it's off if the patch window is hidden).
 - c) Now, place additional trackers on different features of the same surface to replace the trackers that you had to turn off.



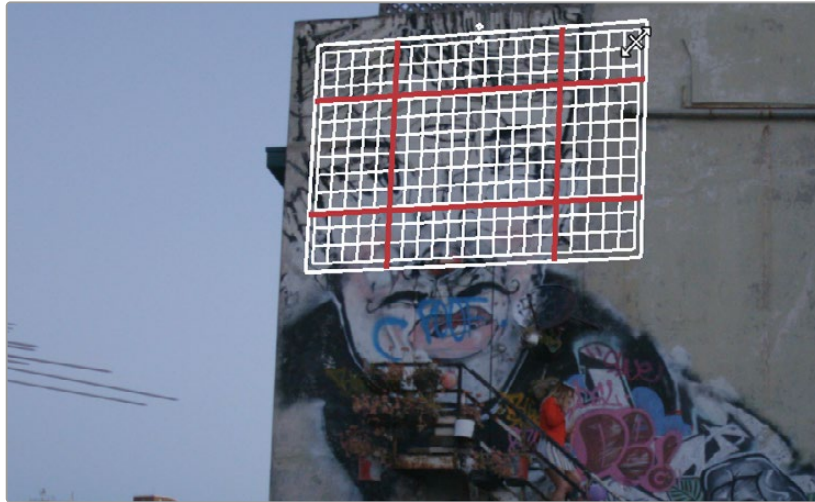
Placing more tracking controls to cover for tracking controls that turn themselves off when going offscreen or being occluded by other moving features in the image

- d) Click "Track forward" again to continue the track, and DaVinci Resolve will automatically use those additional features to continue the analysis.
 - e) When the analysis is complete, turn the trackers you disabled back on again. All trackers that are turned on contribute data to the match move, for whichever ranges of frames they contain valid tracking data.
- 8 When DaVinci Resolve finishes its analysis, move the playhead back to the first frame of the track, and choose Positioning from the "Show controls for" pop-up menu. This takes you to the next stage in the Match Moving process.



Choosing the Positioning controls

- 9 A grid appears, covering the Viewer. This is the positioning canvas, and it lets you corner-pin the area in which the foreground image you connected in step 3 will be composited and tracked. You can corner-pin and resize it by dragging it by the corners, and you can move it by dragging the center.



Resizing the canvas to fit the foreground image into the background

- 10 Once you've positioned the grid, scrub back and forth to verify that it tracks correctly and looks right once it's in motion.
- 11 Choose Compositing from the "Show controls for" pop-up menu to expose the compositing controls.
- 12 Set the Output pop-up menu to "Composite" to output the final match-moved composite. If necessary, you can use the Composite Type and Plate Cropping controls to choose how the foreground image is blended with the background and whether you need to crop unwanted parts of the image from the edges.



The final match moved composite

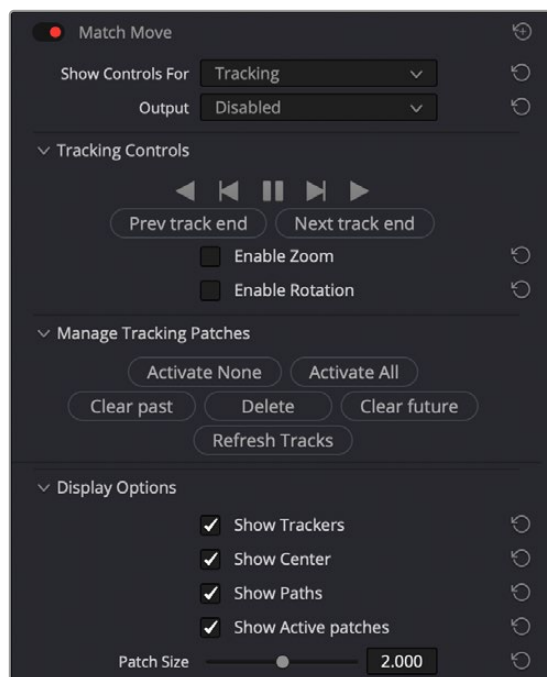
At this point, you've finished the match move.

Main Controls

The “Show controls for” pop-up menu lets you choose one of four pages of controls that take you through the match moving workflow. The Output pop-up menu lets you choose what you want to output, with choices for Disabled, Positioning Reference, and Composite.

Tracking Controls

The Tracking Controls govern the first stage of the match moving process, and this page of controls provides everything you need for setting up, executing, and refining multi-patch motion tracks.



The controls that are exposed when Main Controls is set to Tracking Controls

The Tracking Controls group

The primary controls used to initiate tracking. These consist of the tracking buttons that let you execute tracks either automatically using the outer Track Forward or Track Backward buttons, or manually using the inner Track Next or Track Previous buttons. The middle button lets you pause tracking.

The Next Track End and Previous Track End let you jump to previous and subsequent tracked sections of frames; in cases where there may be gaps in your tracking data, these controls let you jump to different segments of contiguous tracking data.

Two checkboxes let you choose whether you want to track and analyze Zoom and/or Rotation. These must be set before you begin tracking.

The Manage Tracking Patches group

The two buttons at the top let you choose which tracking patches are enabled to contribute to the overall match moving result, or disabled. Activate None and Deactivate All let you turn all patches on or off at once.

Clear Past and Clear Future let you clear bad tracking data forward or back in time from the playhead's current position, for enabled tracker controls. Delete Tracking lets you delete the tracking data at the current frame only for enabled tracker controls.

Refresh Tracks updates a tracker control's patch when the feature it's tracking changes shape, color, or lighting enough to interrupt the track. Select one or more trackers and click Place New Patch in Track to update the tracked feature at that frame.

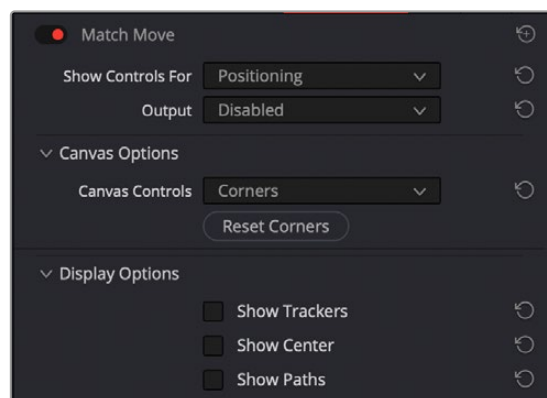
The Display Options group

This group lets you turn the visibility of different onscreen controls off and on. Checkboxes let you Show Trackers, Show Center (the red tracking center point), Show Paths (the tracked motion paths), and Show Active Patches. A slider lets you adjust the size of each patch window.

Show Comp Result lets you see the composited foreground layer against the background layer.

Positioning

The Positioning controls govern the second stage of the match moving process. They provide everything you need to transform the foreground image to fit against the background by moving, corner-pinning, and resizing.



The controls that are exposed when Main Controls is set to Positioning

Canvas Options

The Canvas Controls pop-up menu lets you switch between corner-pin controls, where dragging the four corners lets you corner-pin, dragging the center lets you reposition, and dragging the top, bottom, and sides lets you resize vertically or horizontally.

A Reset Corners button lets you reset the grid to full screen, if you need to scrap what you're doing and try again.

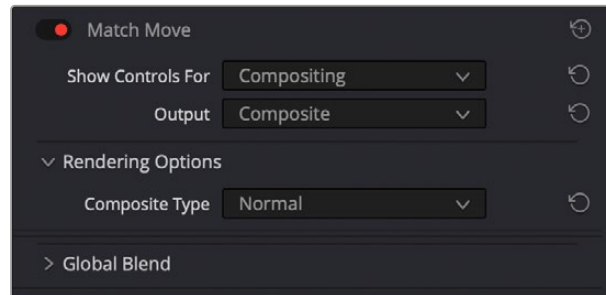
The Display Options group

This group lets you turn the visibility of different onscreen controls off and on. Checkboxes let you Show Trackers, Show Center (the red tracking center point), and Show Paths (the tracked motion paths).

Show Comp Result lets you see the composited foreground layer against the background layer.

Compositing

The Compositing controls govern the third stage of the match moving process, to use when fitting the foreground image against the background image to make a seamless composite. Note, if you need to color correct the foreground image to match the background, you'll need to add a corrector node between the Ext. Matte that's routing in the foreground image into the Color page and the FX node that's doing the compositing.



The controls that are exposed when Main Controls is set to Compositing

Rendering Options

The Composite Type pop-up menu lets you choose whether the composite that's being output is an overlay of the Ext. Matte over the background, or the background Plate only.

Plate Cropping

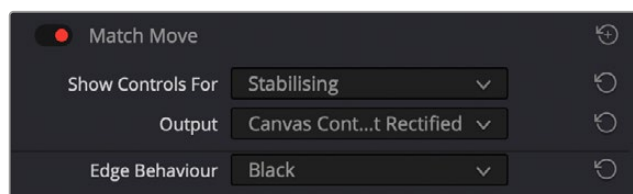
Four sliders let you crop the Left, Right, Top, and Bottom of the image, if necessary.

Global Blend

The Blend slider, present in every Resolve FX plugin in DaVinci Resolve, lets you adjust the opacity of the effect, effectively controlling the opacity of the foreground layer.

Stabilizing

The Stabilizing controls actually have nothing to do with match moving, but instead allow you to use the motion tracking accomplished with the Tracking Controls to stabilize the image. An Edge Behavior pop-up menu lets you choose how to handle blanking around the edges as a result of the background image being transformed to keep the subject of the frame in place.



The controls that are exposed when Main Controls is set to Stabilizing

Surface Tracker (Studio Version Only)

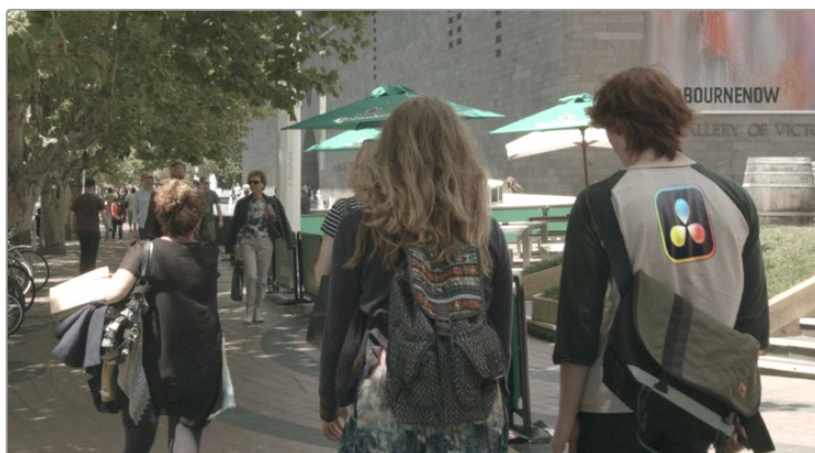
The Surface Tracker is used for applying textures and effects to flexible or deformable moving surfaces, like cloth and skin. The Surface Tracker ensures that the folding and deforming of the texture or effect is realistically matched to the folding and deforming of the surface that it is attached to in the image.

The Surface tracker is a complex effect, with many different options and controls to create the most realistic composites possible. But in all cases you simply move from left to right, using the buttons in the effect's settings.

- 1 Click the Bounds button, and click on the screen to define the area on the surface where you wish to apply the new texture. The bounds should select one surface at a time and not go over/on object edges.
- 2 Click on the Mesh button to define the control points along the natural folds and curves of the surface within the bounds.
- 3 Click on the Track button to analyze the motion of the surface over time, using the Mesh you just set.
- 4 Click on the Result button to composite the texture on top of the moving Mesh.

It is recommended that you use the Surface Tracker as a separate FX Node (dragged from the effects library directly into the node tree), rather than applied to a normal corrector node. This allows you to have access to an additional Alpha input for applying textures that have some transparency built in to them.

Across all the tabs in the Surface Tracker, the Tracked Range shows the start and end frame numbers of your clip that contain tracking data. Any existing tracking data will be reset if you modify the bounds, or mesh layout, and you will need to re-track it.



(Previous Page) The Surface Tracker in action, (Above) overlaying the logo, and warping and conforming it realistically to the folds of the shirt.

Bounds

This is where you are prompted to draw an area on the image that you want to apply the texture to. Ideally, you want to choose a frame in the clip that has the surface you want to track as close and as flat-on to the camera as possible. This frame becomes the default reference frame for the effect. Make sure that the Open FX Overlay controls are active in the Viewer to use the tool. Simply click on the image to make multiple points around the area you want to define (a minimum of three points are required), and the effect will connect them automatically as a polygon. Boundary points should be **on** the surface for tracking, not around it. If points are accidentally placed on object edges or the background instead, it will cause errors in the tracking.

You may add multiple boundaries and multiple holes to the same image, and all boundaries will join together into one shape before subtracting the holes. The purpose of multiple boundaries and holes are to allow you to mask out complex surfaces with regions not for tracking. For example, you can make a boundary around someone's face, but use multiple holes to exclude the eyes and mouth as those change appearance quickly as the subject blinks and talks, making them exceptionally difficult to track.

Multiple boundaries are designed to work on the same surface; using multiple boundaries on different surfaces within the image will cause poor results. Separate surfaces are best tracked by using multiple instances of the Surface Tracker instead.

The boundary is not useful for rotoscoping objects, only for analyzing the motion of a surface. Rotoscoping of foreground objects can be done as described in the Tracking Using a Mask section below.

Click to Add Points: These are the controls used to make boundary areas on the image.

- **Boundary:** The currently selected boundary area on the image. You can change the current boundary or hole by using the drop-down menu. If there are issues with the boundary (i.e., crossed edges, too few points, etc.), an error message will appear next to the boundary name.
- **+ Bound:** Adds a boundary polygon to the image. You can have multiple boundaries defined for the same surface, and they will be joined together into one shape.
- **+ Hole:** Adds an area to exclude tracking from the surface. Used to remove overly complex and changing parts of a surface from the effect's calculations.
- **Delete:** Lets you delete the boundary or hole selected in the Boundary field. Currently, this operation is not undo-able, so be careful around this button.
- **Clear All:** Lets you delete all boundaries and holes in the effect.

Mesh

With the boundaries now defined, you can establish the mesh over the surface to be tracked. The initial mesh serves as the starting point for tracking, so for best results the mesh should be placed when your surface is as flat-on and closest to the camera as possible.

The mesh can be manually adjusted by dragging current points (multiple points can be moved by holding command and dragging an area around the points) or clicking on an area to place new ones. Option-clicking a point will remove it from the mesh. Manually adding mesh points can be useful if there are flexible portions of the surface known to be a point where the texture should fold but which is not initially identified when placing the mesh. For example, an extended arm may not have a mesh

point active at the elbow, but with a mouse click (and a basic knowledge of human anatomy), you can manually add one there. Another use case would be to add points to an area with little detail, but you know will develop detail as it rotates into view. Placing mesh points in blank areas that show little detail will not improve results and may actually degrade them, so very few points on smooth areas is desirable. A good rule of thumb is that if more than five points are created inside the boundaries, the mesh may be good enough for tracking. Points cannot be added or deleted after tracking.

A good mesh is a sign of good tracking as they go hand in hand. If your mesh has few points detected and the results don't track tightly, you can artificially improve your results. For example, make sure that the content is clearly visible. Tracking and point detection can fail if the surface is too bright (HDR), too dark, or low contrast due to working in log. To improve results, you can add a corrector node before the Surface Tracker to adjust contrast and levels manually, then run the Surface Tracker. You could also use a corrector node with the Contrast Pop effect before the Surface Tracker to help track surfaces that are too smooth. In both cases simply disable the corrector node again after the tracking has finished.

- **Regenerate Mesh:** This button recreates the mesh at the current frame and settings and discards any manual changes that have been made.
- **Point Locations:** These controls determine which mode is used to generate the mesh points.

Automatic: The Surface Tracker automatically locates the best details for tracking, and places mesh points on those details. In most cases Automatic will give you the best results.

Point Number Limit: Lets you set the maximum amount of mesh points to place. The number of mesh points affects how the overlay will flex and fold, and should be set by the expected flexibility of the surface. They do not directly affect tracking speed.

Minimum Point Spacing: Lets you set the minimum distance between mesh points. This prevents mesh points from bunching together too tightly in highly detailed areas and interfering with each other.

- **Uniform Grid:** Creates a regular grid of points in the boundary and relies on internal shape models to deform the grid during tracking. This may give better results than Automatic mode if your surface is highly detailed.

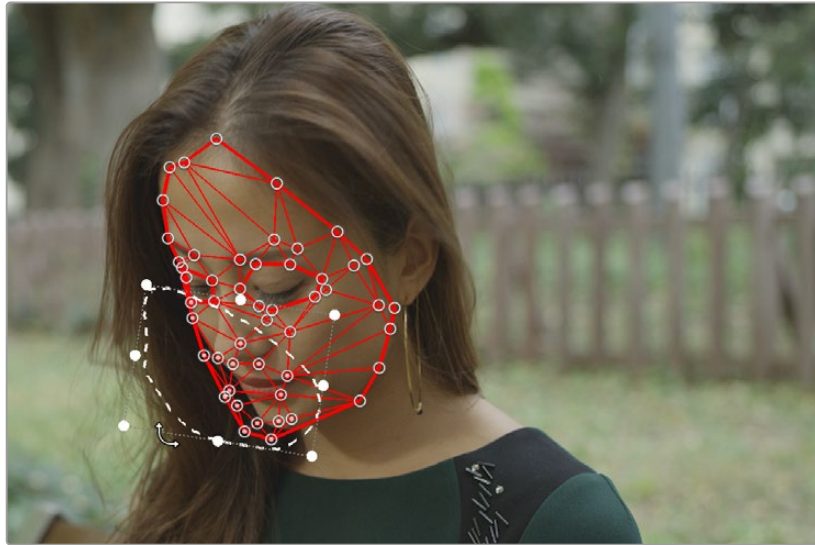
Horizontal Spacing: Adjusts the horizontal distance between adjacent points in the uniform mesh.

Vertical Spacing: Adjusts the vertical distance between adjacent points in the uniform mesh.

Track

This section of the effect tracks the motion of the mesh points over time. Use the Tracking Direction controls to complete the tracking procedure across the clip. After the tracking is complete, the effect does no more analysis, and any additional contrast assistance from previous nodes can be discarded after this step. For simple cases, the tracking may run completely without user intervention, but for more complex cases, a number of additional controls are used to refine the track.

Fixing a broken track will require some manual adjustments to both the mesh and the tracking data. At any time while tracking, mesh points can be adjusted either one by one, or with the lasso selection tool (Command - Drag) around the entire mesh or just parts of it. The lasso selection tool also has the ability to stretch, scale, and rotate the mesh points, making it easier to manually deform and place the mesh exactly where you want it to be, if for some reason the automatic tracking fails. However, additional mesh points cannot be added or deleted at this stage, without first resetting the tracking.



Command-Drag will allow you to lasso multiple mesh points, and scale, rotate, and stretch them, as needed.

Manually editing the mesh points at a frame only affects that frame directly; it does not affect neighboring tracking data. However, it does affect neighboring interpolated positions, which makes it good for tracking from that point onwards. For example:

- Tracking is good until frame 150 when a few points separate from the rest, disturbed by a lighting change. At frame 150 they can be manually moved back into place, then if tracking is resumed they will continue with their corrected positions.
- Tracking exists until frame 150, after which there is no data for a few frames until it resumes at 160. Editing the points at either frames 150 or 160 will cause the intermediate frames to be interpolated with these changes applied.
- Tracking exists up to frame 150 and at frame 160 onwards, frames 151-159 are interpolated. At frame 155 some points are adjusted. This creates tracking data at this frame, and now frames 151-154 interpolate from the last tracked frame to this current one, and then frames 156-159 from the current to the next tracked frame.

If an object in the frame occludes the tracked surface, for example tracking a mesh on a shirt, and the person's arm moves down through the mesh, breaking the track. You may be able to fix this issue by using an Alpha channel, as described in the Tracking Using a Mask section below.

Tracking data is also visible as keyframes in the Keyframe Editor in the FX node's Surface Tracker keyframing data, and it is also accessible as the last control at the bottom of the Track section where the data at a frame can be created or deleted directly.

- **Go to Start:** This button will snap the playhead to the first frame that has been tracked, or the frame where the boundary was created.
- **Go To End:** This button will snap the playhead to the last frame that has been tracked.
- **Tracking Direction Controls:** These controls perform the tracking analysis in the direction indicated. They are from left to right: Track Backwards, Track Backward One Frame, Pause Tracking, Track Forwards Then Reverse, Track Forwards One Frame, Track Forwards. Using these controls, you can set the track across the whole clip or limit the tracking to just a section of it.
- **Tracking Data Controls:** These controls allow you to manipulate the track data after it's been run.
Remove Tracking Data Before This Frame: Clears all the tracking data before the current frame.

Track From Last Known Match: This button will match the tracking data to the current frame and connect it to the last frame where tracking data exists. If there has been too much motion between those times, you can move the points approximately into place and try matching again. For example, the tracking occurs without issue until frame 150, but something moving across the scene causes an error in the track for six frames (151-157). The tracking data can be paused before it hits these frames, or these frame's tracking data can be deleted after the fact. Then at frame 158 you can press this button to continue the track from the last known good track at frame 150.

Remove Tracking Data At This Frame: Clears the tracking data at the current frame.

Remove Tracking Data After This Frame: Clears all the tracking data after the current frame.

- **Interaction Overlays:** These buttons control the visibility of the mesh overlays on the Viewer.

Show: Shows the mesh at all times.

Hide on Drag: Shows the mesh until a point is dragged, then it is hidden, allowing you to see the surface underneath for fine adjustments.

Hide: Hides the mesh at all times.

- **Tracking Behavior:** These options allow you adjust the way the tracking operates.

Motion Range: Controls how far to search for matches. Increase this slider if the surface is moving too quickly and the tracking fails to find it.

Mesh Rigidity: Controls how flexible the mesh is. Reduce this slider to make the mesh points more flexible. This gives you best results if the clip is noise-free and warping smoothly. Increase this slider to get a “stiffer” tracked surface with less internal warping.

Quality: The tracker can be very computationally intensive; it is recommended to use the Faster setting while making adjustments to the mesh, then use Better for the final track once all the parameters have been set correctly.

Tracking Keyframes: Controls the Surface Tracker in the standard Keyframe Editor. You can insert, delete, and navigate different keyframes from this option.

NOTE: The Surface Tracker creates data at each frame of the source clip. If the source clip is not at the same frame rate as the timeline, tracking may seem to stutter or skip frames. You can “bake” retiming into a clip by right-clicking on it on the Edit page and choosing “Render In Place” to create a result with exactly one frame for every timeline frame.

Result

Once the tracking data is complete, the Result section lets you determine what and how to composite a video layer on top of the moving surface.

- **Output:** There are six options for what to do with the resulting tracking data.

Warp Input 2 Onto 1: Warps the Surface Tracker's second RGB input to the shape and position of the tracked mesh and composites it onto the clip.

Warp Input 2 Onto Blank: Warps the Surface Tracker's second RGB input to the shape and position of the tracked mesh and passes the Alpha channel information, allowing you to perform the actual composite in another node or effect.

Warp Input 1's Alpha: Applies the results of the Surface Tracker to the first Key input (the second RGB and Key inputs are ignored). If the first Key input receives an Alpha channel from another node with, say, a drawn shape, or a key, or other Alpha data, it can be warped from the first frame to others.

Create Mesh Alpha Mask: Produces an Alpha mask, which follows the position and shape of the outline of the mesh. All inputs and alphas are ignored.

Stabilize-Warp Input 1: Performs an inverse warp to the Reference Frame (again, this should be a frame that ideally is as close to and flat-on to the camera as possible). The result is a stationary image within the boundary with all the lighting changes and details of the surface's motion shown on it. This now stabilized image can be modified with any of DaVinci Resolve's existing tools, bypassing any issues caused by this area's motion in the original video clip. Once modified, you warp the result back onto the original footage by using a copy of the original Surface Tracker node but with the Rewarp Stabilized Clip setting here instead.

Rewarp Stabilized Clip: Composites the output of the Stabilize-Warp Input 1 selection back onto the original video clip, after it's been modified. This option is designed to be used in conjunction with a copy of a Surface node using Stabilize-Warp Input 1. It seamlessly blends whatever additional modifications that were made between the two Surface Trackers back together.

- **Overlay/Alpha Source:** Allows you to select which frames of the overlay or alpha are used to create the result.

First Frame: Useful for freezing content from an overlay clip to use a still frame as a static texture. In DaVinci Resolve's External Matte controls the offset can be chosen, but also you can use the same frame as the texture reference with Reference Frame.

Reference Frame: Starts at the currently defined reference frame (see below), instead of at frame 0. The reference frame is the frame which the mesh overlay is initially positioned in its "un-warped" state, and the positioning controls are active.

Each Frame: Treats the overlay or alpha as a clip, warping each frame in turn. This is necessary for animated overlays or re-warping stabilized content. It should be noted that in the Key controls on the Color page, when an External Matte is selected, options appear to control the lifetime of the external clip.

- **Expand/Contract:** Allows you to add or remove pixels from the edges of the overlay or alpha. It is important to note that the expanded area is not tracked and will not be stabilized to the content.
- **Softness:** Blurs the edge of the Expand/Contract tool to blend more realistically into the scene.
- **Motion Blur:** This slider lets you add motion blur to the overlay to preserve the illusion that it was actually captured with the scene.
- **Overlay Placement:** These controls let you set the initial transform options for the image you want to overlay on the mesh.

Go To Reference: Moves the playhead to the selected reference frame.

Set to Current: Sets the current frame under the playhead as the reference frame.

Reference Frame: Shows the frame number at which the overlay is originally positioned and un-warped (the point from which the warp to every other frame is calculated). The positioning controls appear on screen when the clip is at the reference frame. A frame where the surface's view is largest and least warped is the best starting point for rendering onto other frames.

Positioning: Lets you choose how to adjust the overlay position.

- **Sliders:** Choosing this option exposes a set of transform sliders (X/Y position, zoom, rotate, etc.) that let you manipulate the overlay's initial position.
- **Interactive Canvas:** Choosing this option exposes draggable frame controls in the Viewer, allowing you to transform the overlay using mouse controls. The Viewers shows a large grid with 9 regions; you can drag in any of them to move, skew, or warp the overlay.
- **Interactive Pins:** Adjusting the image in this mode is done by manually placing control points, called pins, in the Timeline Viewer. Adding one pin only gives you position control. At least two points are required for scaling and rotation. Dragging on one of the pins scales or rotates the image around the other pin. Using three pins, you can create perspective distortions by dragging any one of the pins. You can add up to four pins for unique corner pinning distortions.

Reset Position: Resets the on-screen overlay back to its default state.

- **Interaction Overlays:** These buttons control the visibility of the mesh overlays on the Viewer.
 - Show:** Shows the mesh at all times.
 - Hide on Drag:** Shows the mesh until a point is dragged, then it is hidden, allowing you to see the surface underneath for fine adjustments.
 - Hide:** Hides the mesh at all times.
- **Compositing:** Lets you set the composite blend mode for the overlay operation. For more detailed information on each blend mode, see *Chapter 50, “Compositing and Transforms in the Timeline.”*
 - Composite Type:** A drop-down menu of all the possible composite operations.
 - Opacity:** This slider controls the transparency of the overlay.

Using the Surface Tracker

The Surface Tracker is a deep and versatile tool that can be used to realistically isolate a moving surface for a variety of effects. The examples below can get you started on discovering its uses.

Using the Surface Tracker to Composite a Logo onto a Fabric Surface

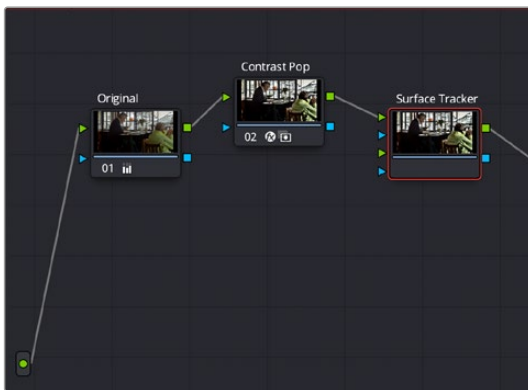
One of the more common use cases of the Surface Tracker is to overlay a still graphic on top of a moving surface and have the graphic behave realistically as the surface deforms and warps. In this example, we will add a simple logo to the back of a jacket.

The first step is to define the boundary of where we want the surface analyzed. For best results, you need to place the boundary on the surface to be tracked, not around it, and try to find a frame where the surface is flat on to the camera and as large in the frame as possible.

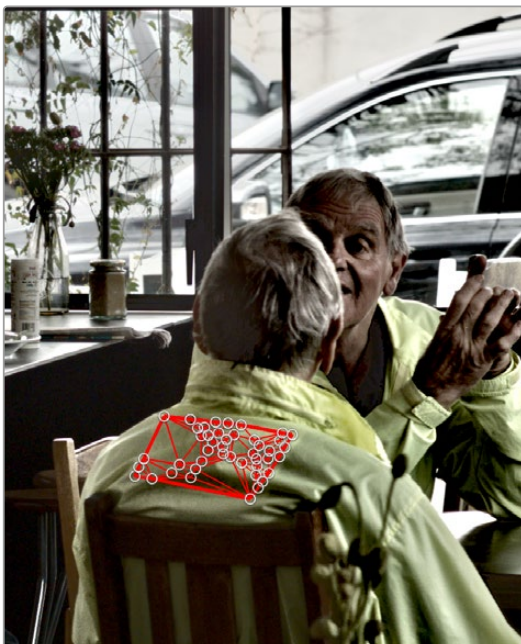


The next step is to define the Mesh within the surface to be warped. In this case, we will leave the Point Locations setting at Automatic to have the effect apply the mesh points. Ideally, there will be more than five mesh points for a good track. Unfortunately, this surface doesn't have enough detail to make a good mesh, so we will temporarily add a Contrast Pop node before the Surface Tracker in order to artificially enhance its details.

Creating the initial boundary to isolate the surface of the jacket where the logo will be applied



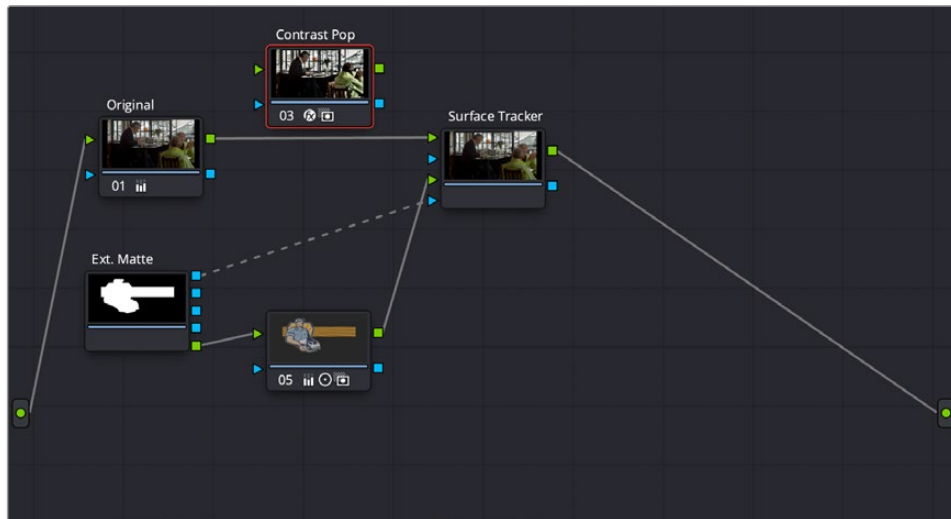
By adding the Contrast Pop FX before the Surface Tracker input, we can artificially increase the detail inside the boundary, giving it a better surface to form the mesh.



Next, we will track the Mesh's motion using the Track Forward button and leave the rest of the controls at their defaults. Note that the track continues to warp and fold correctly, even when the subject leans back in his chair.

The Tracker analyzes the Mesh movement throughout the rest of the video clip.

With the surface now correctly analyzed, we no longer need the Contrast Pop node, so we can disconnect it at this point. Next, we will drag the logo (a .png file with transparency) from the Media Pool, over into the node tree, where it comes in as a matte. We then add a corrector node after the matte, so we can adjust the color/saturation/brightness, etc. of the logo. Next, we set the RGB and Key connections of the nodes through the matte and corrector nodes, and into the second RGB and Key inputs of the Surface Tracker. This is shown on the node tree below:



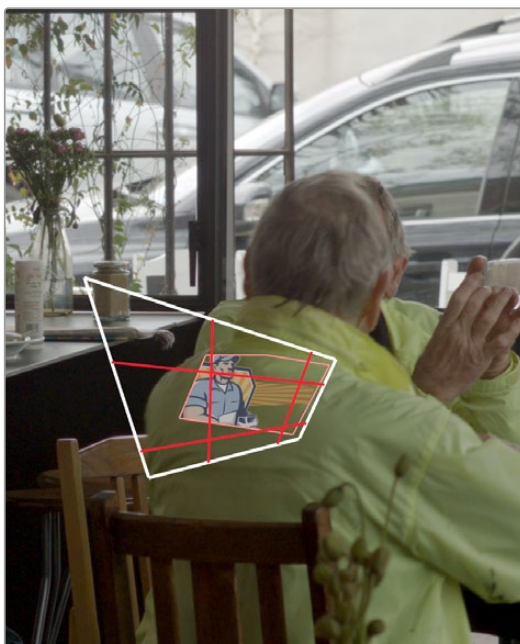
The node tree, showing the layout of the RGB and Key inputs to composite the logo over the jacket

Now, moving to the Result section, we set the following parameter:

Output: Warp Input 2 onto 1

This will take the second RGB and Key input (logo) and deform it over the original image coming in to RGB input 1.

Then we will click on the Go To Reference button to take the clip to its reference frame, and in Positioning, use the Interactive - Canvas tool to reframe the logo to look like it's on the same plane as the jacket.



Adjusting the logo in the Interactive Canvas mode to transform it to make it appear it's in the same plane as the jacket

For the last step, we change the Composite Type from Normal to Overlay to make the logo blend in better to the surface of the jacket. Notice how the logo is deformed by the mesh to wrap around the curves and folds of the jacket, and it tracks along with the man as he leans back in his chair.

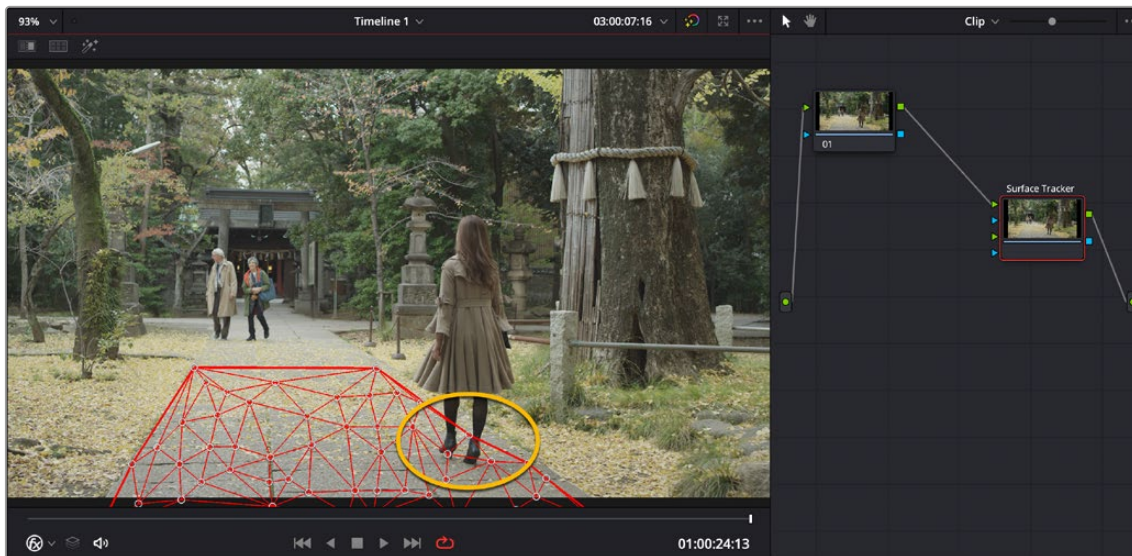


The finished Surface Tracker composite; the logo now adheres to the curves and folds of the jacket and tracks along with the man as he shifts positions in his chair.

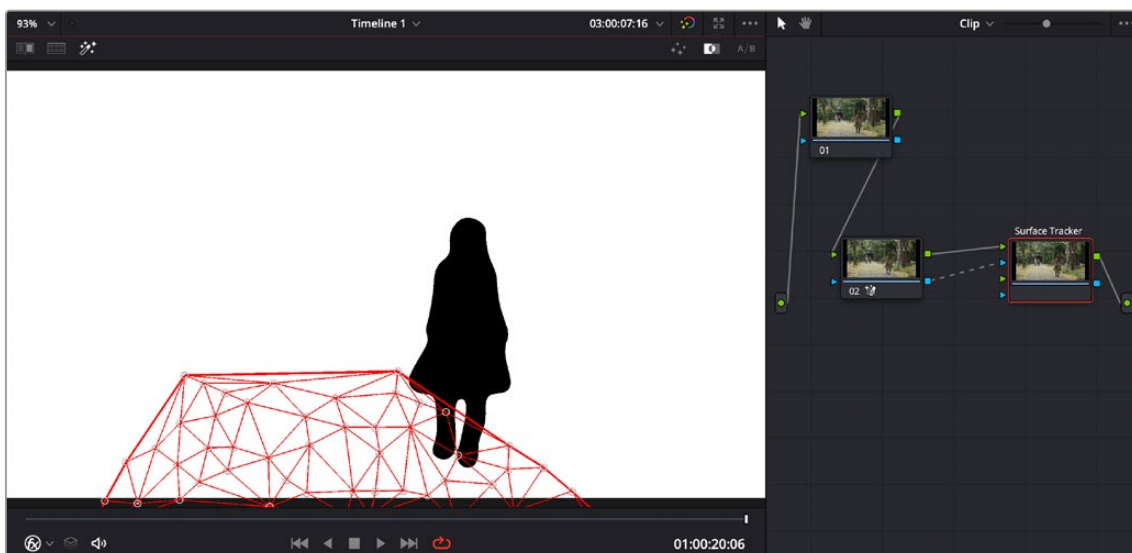


Tracking Using a Mask in the Surface Tracker

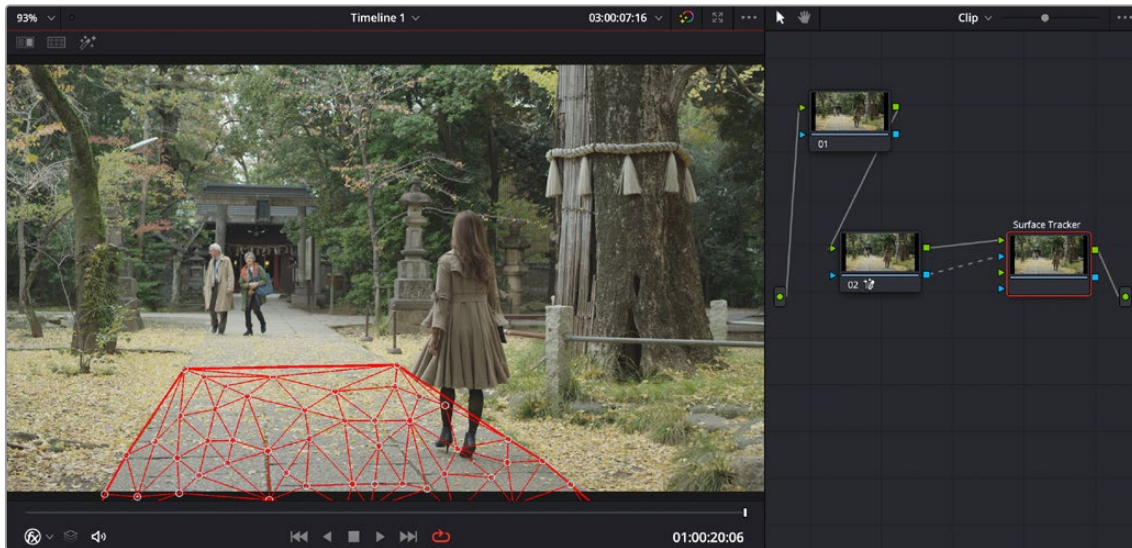
If there is an object occluding the surface to be tracked, then an input key can be used as a tracking mask to exclude the blocking object from the track. Areas of full alpha (completely white) in the Alpha channel are analyzed for tracking, while other areas are excluded. Any appropriate effect, window, or qualifier can be used as the Alpha channel generator. For example, in the shot below, the path has the surface mesh applied to it, however the woman walking over the path breaks the mesh tracking. By connecting a node's key output with a Magic Mask applied to it that picks out the shape of the walking woman, then inverting that mask (so the woman is black, while her surroundings are white), the mesh ignores her legs and feet as they pass through it and only tracks points in the white areas (everything other than the woman).



A problem trying to track the path's surface; as the woman walks through the mesh, the points attach themselves to her feet instead of the path, breaking the tracking on the right side of the path.



By connecting an inverted key from a Magic Mask node of just the woman, the tracking ignores the woman walking through the mesh (anything black), and only tracks points in the white part of the Alpha channel.



The resulting mesh is now properly attached to the path's right side and tracks correctly as she walks through it.

In the final step of the Surface Tracker, the overlay with alpha is composited over the original clip with alpha, so if using a tracking mask, it is recommended to disconnect it from the Surface Tracker's key input after the tracking is complete.

Using the Surface Tracker's Stabilize Warp

The Surface Tracker's Stabilize Warp operation allows you to use the mesh to freeze a section of video in place to make it easier to use with certain other operations. This is useful when the surface you are modifying is difficult to isolate with other tools due to its movement. Using the Surface Tracker to stabilize areas is a bit different to set up.

- Initially you want to set the Surface Tracker's mesh tracking on a surface you want to work on, as explained earlier in this chapter.
- Once complete, in the Result section you use the Output mode called Stabilize-Warp Input 1, and then feed that RGB output to a new corrector or FX node.
- This will isolate just the mesh surface and freeze it in place, allowing you to make any adjustments needed. This method is different than, say, a window qualifier, which sticks to the video and moves with it. In Stabilize Warp, the "window" (actually the mesh defined boundary) is frozen in place instead, and the video moves and warps within it.
- Once your adjustments are finished, you then feed the RGB output of the corrector node back into an exact copy of the first Surface Tracker node. The copied Surface Tracker node's Output then needs to be set to Rewarp Stabilized Clip instead. This re-lays the adjusted mesh back on to the original video clip. Directly copying an FX node like the Surface Tracker is not currently supported. To copy the Surface Tracker, you must grab a still into your Gallery, then right-click on it and append the still's node graph back into your same clip, deleting the excess duplicate nodes other than the Surface Tracker.

For example, in the clip below we will use the Surface Tracker to remove a small mole from the woman's chin. Normally, we could use a Patch Replacer FX to put a smooth area of skin nearby over the mole. This shot is tricky to use effectively in the Patch Replacer because the subject cocks her head while she's talking, as the camera trucks around her.

This movement on all axes means that the area of skin underneath the source patch is changing angle and lighting all the time, ruining the effect. While the patch replacer does

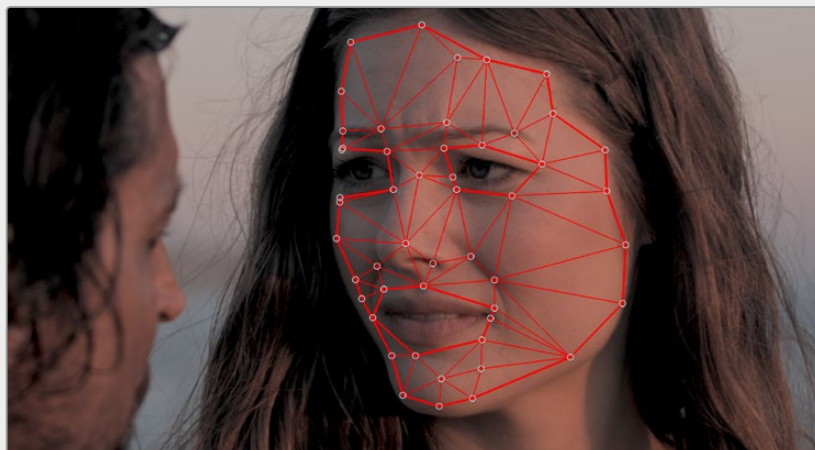
support tracking, the mole is on her chin and her talking also causes the skin around her cheek to deform, making it unsuitable for a simple track.

In this case, we can use the Surface Tracker's Stabilize Warp function to "hold" those face area pixels in place under the source and target patches, patch them, and then make a Rewarp Stabilized Clip version of the Surface Tracker to lay those corrected pixels back onto the original image.



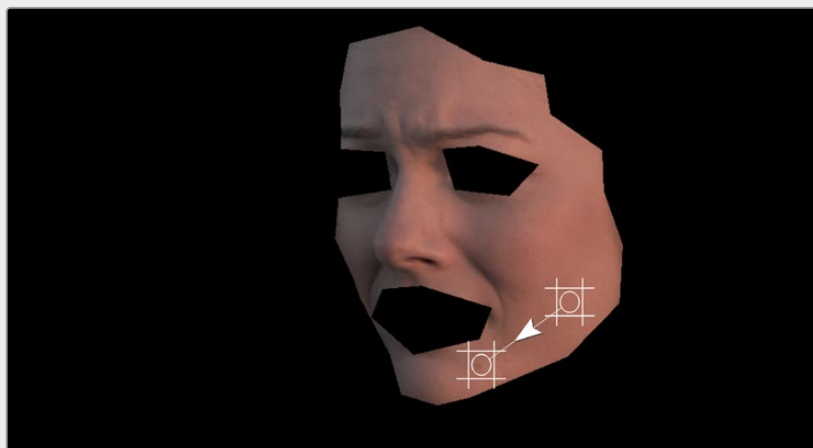
We want to remove this mole from the woman's chin. Because of camera movement, combined with her chin and cheek deforming as she talks, this is difficult to achieve with the Patch Replacer alone.

First we will create a mesh on her face and track it as described above. Making holes for the eyes and mouth so their rapid movement does not disturb the mesh on her face.



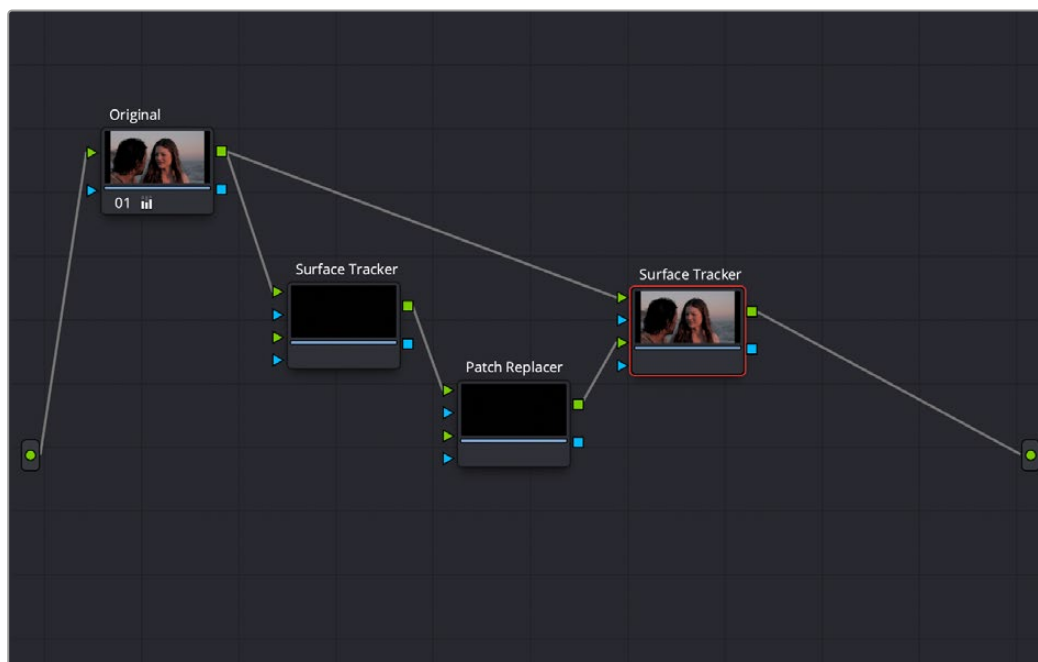
We apply the mesh to her face and track it, leaving holes in the mesh for her eyes and mouth.

Once the tracking is complete, in Result we will choose the Output mode Stabilize-Warp Input 1. This will freeze the mesh in place and give us the resulting output, which we will feed into a Patch Replacer node. Now we can simply use the Patch replacer statically to hide the mole using another area of skin. No tracking is required. Because the mesh is static, and the video warps within it, this area now perfectly matches the angles, lighting changes, and skin stretching of the area as she talks.



The frozen mesh stays in place (the face) and the video warps inside it. This allows us to use a static Patch Replacer to replace the mole area, with a smooth area on her cheek. No additional tracking is required.

At this point, we will then make an exact copy the Surface Tracker node, and place it after the Patch Replacer. To copy the Surface Tracker node, we must grab a still of the current clip, and then right-click on the still in the Gallery and select Append Node Graph to duplicate it back into our node tree. We want to delete any excess nodes added, and keep only the copy of the Surface Tracker. Then in the copied Surface Tracker, we need to select “Rewarp Stabilized Clip” from the Result tab’s Output mode. Then connect the original clip to RGB input 1, and the output from the Patch Replacer to RGB input 2, as shown in the node graph below:



The node tree showing the correct setup for Warp Stabilizing the mesh. The first Surface Tracker node has its Result Output set to “Stabilize-Warp Input 1.” The second copy of the Surface Tracker node has its Result Output set to “Rewarp Stabilized Clip.” The actual mole removal will take place in the Patch Replacer FX node.

Our finished effect now shows that the mole has been replaced, and regardless of her jaw movement or the camera position, the source patch always matches perfectly to the lighting and position changes over time. This makes for a much more seamless and convincing effect.



The resulting shot perfectly hides the mole (above), no matter her facial position or camera movement.

Transform

The Transform Resolve FX adds considerable control over the standard transforms in the Inspector's Sizing or the Color page Sizing palette. This effect includes all the standard Position, Tilt, Zoom, and Rotate controls but adds corner pinning, two kinds of on-screen controls, motion blur, and Edge behavior controls.

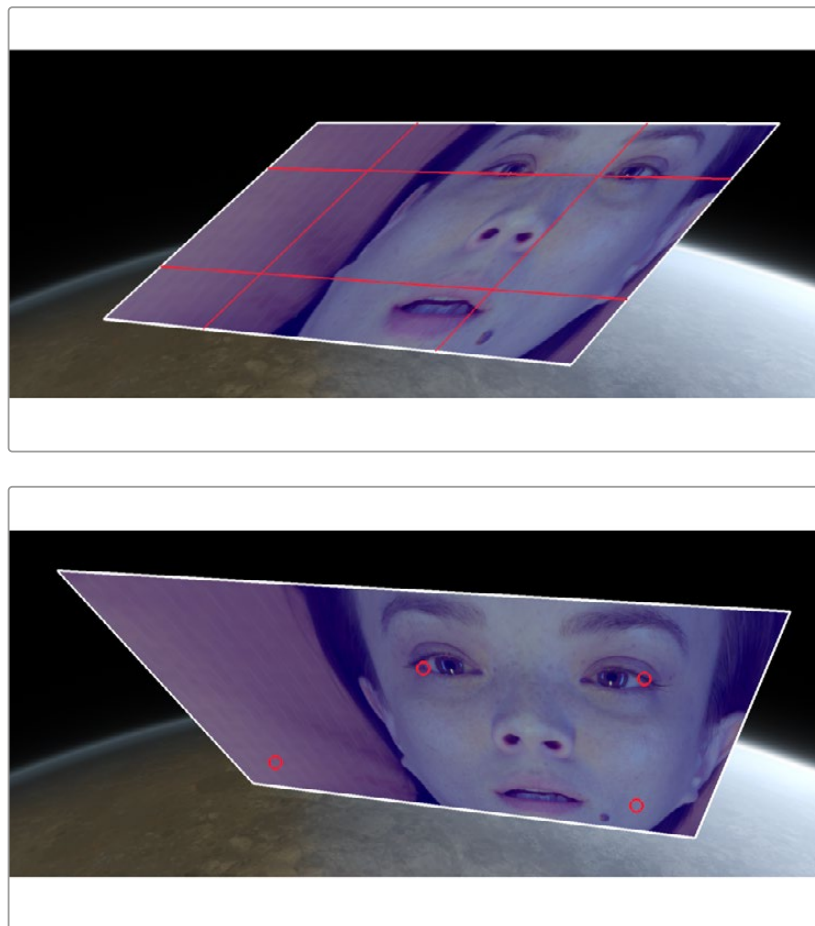
General

- **Control Mode:** This drop-down menu includes three methods for controlling the transforms. One is based on sliders in the Inspector, while the other two are based on manipulating the Viewer's image.
Sliders: Uses sliders and checkboxes in the Inspector to control the transforms. This method has no Viewer overlays with which to control the position, scale, or rotation but provides exact control using numeric values.

Interactive-Canvas: Choosing this mode moves the controls from the Inspector to the Timeline Viewer. The Viewer's controls are shown as a bold white outline and four vertices, one at each corner. You can drag on any part of the white outline for stretching, squeezing, and corner pinning. Dragging the center of the image repositions the frame. The red subdivision overlays make it easy to create shearing effects by dragging line segments to move two corners simultaneously.

Interactive-Pins: Adjusting the image in this mode is done by manually placing control points, called pins, in the Timeline Viewer. Adding one pin only gives you position control. At least two points are required for scaling and rotation. Dragging on one of the pins scales or rotates the image around the other pin. Using three pins, you can create perspective distortions by dragging any one of the pins. You can add up to four pins for unique corner pinning distortions.

TIP: You can Option-Click on a pin to remove it or use the Reset Canvas button to remove all the pins.



Resolve FX Transform Interactive-Canvas mode (Top) and Interactive-Pins mode (Bottom)

Only one Control mode can be active at a time. For instance, you cannot make adjustments with the sliders and then switch to one of the interactive methods. The sliders will be reset once a different method is selected. However, you can apply two Transform effects to a clip, adjusting one with Sliders and the other using an interactive method.

Position Controls

- **Canvas Keyframe:** Only displayed when the Control mode is set to Interactive-Canvas or Interactive-Pins. The Canvas Keyframe has no control except for a keyframe button allowing you to keyframe the changes you make in the Viewer.
- **Reset Canvas:** Resets any interactive changes made in the Viewer and any keyframes created from those changes.

The following parameters are only displayed when the Control Mode is set to Sliders.

- **Position X and Y:** Moves the image within the frame, allowing pan and scan adjustments to be made. X moves the image left or right, and Y moves the image up or down.
- **Zoom:** Allows you to blow the image up or shrink it down.
- **Rotate:** Rotates the image around the center.
- **Width:** Stretches or squeezes the image in one direction only.
- **Height:** Stretches or squeezes the image in one direction only.
- **Pitch:** Rotates the image toward or away from the camera along an axis running through the center of the image, from left to right. Positive values push the top of the image away and bring the bottom of the image forward. Negative values bring the top of the image forward and push the bottom of the image away. Higher values stretch the image more extremely.
- **Yaw:** Rotates the image toward or away from the camera along an axis running through the center of the image from top to bottom. Positive values bring the left of the image forward and push the right of the image away. Negative values push the left of the image away and bring the right of the image forward. Higher values stretch the image more extremely.
- **Flip Horizontal/Vertical:** Two checkboxes let you flip the image in different dimensions.
Flip Horizontal: Reverses the image along the X-axis, left to right.
Flip Vertical: Reverses the clip along the Y-axis, turning it upside down.

Image Adjustment

- **Crop:** This checkbox exposes the crop tools, allowing you to selectively remove parts of the frame from each side.
- **Edge Softness:** This control sets the amount of feathering to the Alpha channel for compositing.
- **Edge Rounding:** This control rounds the corners of the Alpha channel for compositing.

Animation

- **Motion Blur:** Keyframing the transform so the frame moves rapidly will create a blurring effect. The amount of the blur is controlled using the Motion Blur slider.

Advanced Options

- **Edge Behavior:** This menu determines how edges of the frame are handled when the transforms are scaled smaller than the Timeline Resolution.

Transparent: The area outside the transformed image is set to be transparent, allowing the video on tracks underneath this one to be shown behind the image.

Reflect: The image is flipped and flopped to create a mirrored image, which extends to the Timeline Resolution frame boundary.

Wrap-Around: Duplicates the image to create a video wall effect, which is used to fill in the space to the Timeline Resolution frame boundary.

Replicate: Duplicates the outermost pixels along the edge of the image. The pixels are stretched out from each side to reach the Timeline Resolution boundary.

- **Composite Type:** Sets the Composite type for blending multiple video layers together.

For more information on what each Composite type does, see *Chapter 50, “Compositing and Transforms in the Timeline.”*

Global Blend

- **Blend:** Lets you dissolve between the image with no Transform applied and one with the Transform fully applied.

Video Collage

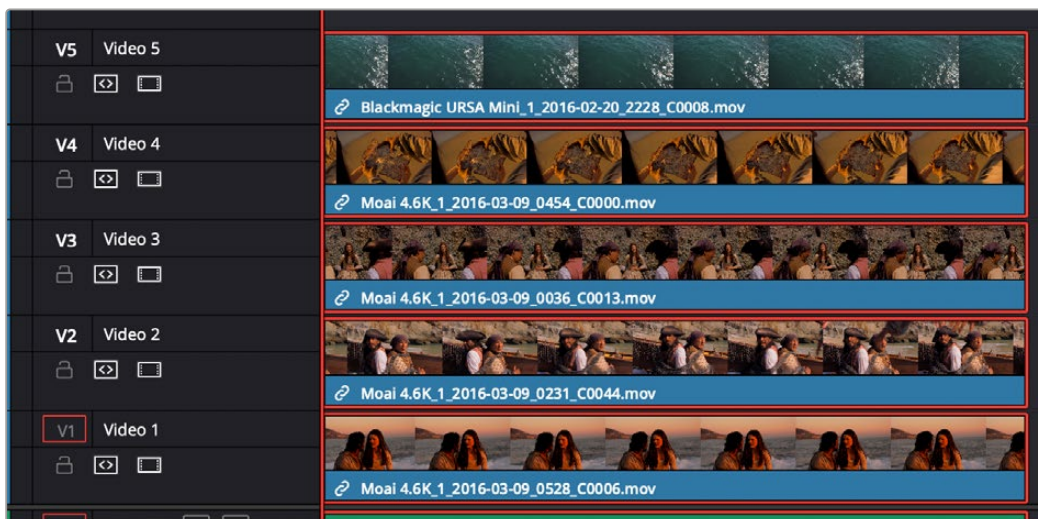
Video Collage is designed to make it easier to create grid-based layouts of multiple video layers to create different kinds of split-screen effects, where each video layer appears within a “tile” that can be styled in different ways. Layout controls let you choose how many rows and columns there are, which determine how many tiles are available to arrange tiles of video next to one another in different ways, with additional controls for creating offsets, changing the margins around and spacing between the tiles, and rounding the corners of each tile.

This effect is designed to be used in the Cut and Edit pages, and there are two ways that you can set this up, which are chosen using the Workflow drop-down menu.

Create Background

The Create Background option is perhaps the simplest way to create a split-screen using this effect. This mode creates a frame with holes in it, behind which you can position multiple layers of video. These clips are placed on tracks underneath the background clip in the Timeline, and use each clip’s individual sizing controls in the Inspector (which can be used via on-screen Transform controls in the Viewer) to scale and position them correctly within the holes. The Video Collage effect is applied to the topmost superimposed video or still image clip in the Timeline, in order to use that clip as a background for the frame.

In the following example, the Video Collage effect is applied to the clip on track 5, which has four other clips underneath it in a stack, each of which has been positioned to fit in one of the holes of the frame.



A stack of clips to create a split-screen effect; the top-most clip is used as the background by the Video Collage effect in Create Background mode, while the stack of clips underneath are all positioned to fit into the holes created in the top clip.

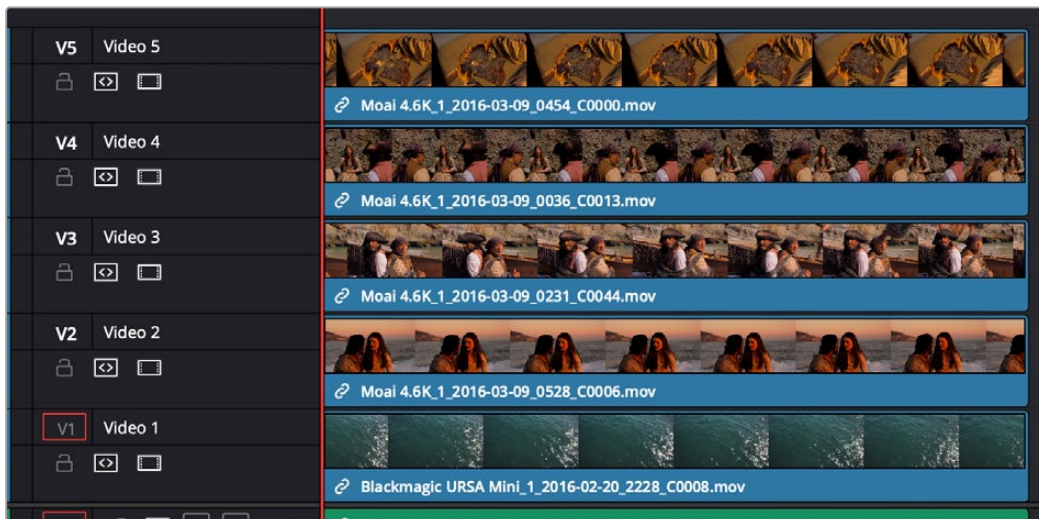
Create Background mode makes it easy to create a split-screen layout of different clips without having to resize them with precision to align the edges and sizes of each clip. The width of the frame that's created gives some wiggle-room to reposition and animate each individual clip within the hole it shows through. You can see the resulting effect created in the previous example below.



The resulting split-screen effect, with four clips arranged to show through the holes created in the top “background” clip that’s creating the frame; you can see the onscreen controls corresponding to the clip on track 4 that have been used to position and resize it to fit.

Create Tile

This mode of operation is a bit more work to set up, as it requires you to create a layout in one instance of this effect, but it gives you more options, automatically transforming each layer of video in your stack of clips in the Timeline to fit a layout you design. The best way to work in Create Tile mode is to edit a stack of superimposed clips in the Timeline, but this time if there’s a clip you want to use as the background, it should go at the bottom of the stack.



A stack of clips to create a split-screen effect in Tile mode; the bottom clip is going to be used as the background.

Then, apply the Video Collage clip to the topmost clip, and set the Workflow drop-down menu to Create Tile. Immediately, you'll see that the current layer has been fit inside the first rounded tile of the default layout of four tiles. For clarity, video tracks 4, 3, and 2 have been muted so you can see the current clip turned into a tile against the bottom clip that serves as the background for this arrangement.



The topmost clip set to create the first tile in a layout; video tracks 4, 3, and 2 are disabled to show the top clip against the bottom clip that's being used as the background.

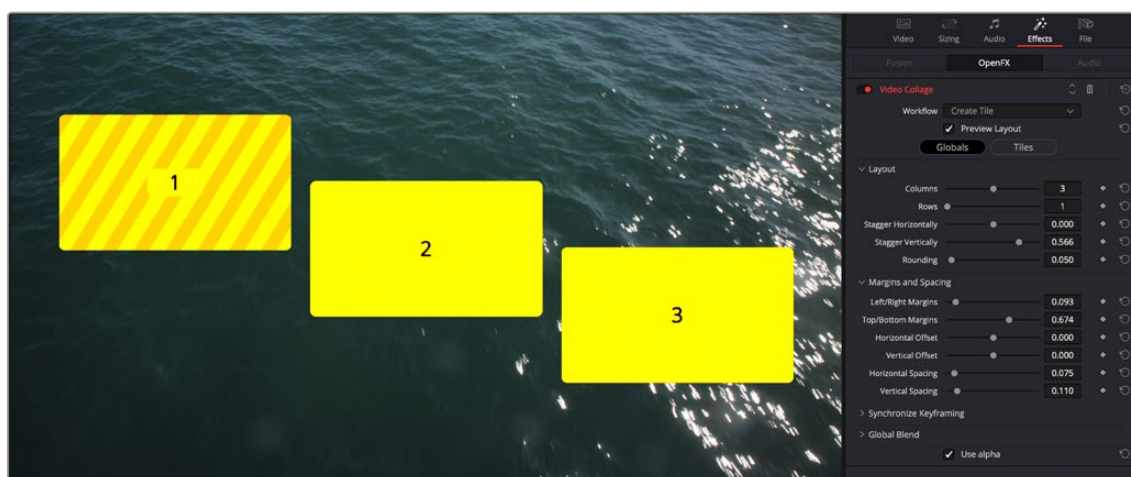
Ultimately, you'll use Paste Attributes to apply a copy of this effect to each clip on the Timeline that's supposed to fit inside a tile, and once you choose which clip goes inside each tile, that's what will create the arrangement. Before you do that, however, you need to customize the layout.

To get a preview of the entire layout for further customization, turn on the Preview Layout checkbox in the Inspector. This draws a preview of how all the tiles will be placed using the current layout. Using this preview, you can make whatever changes you want to set things up, before you copy this plugin to all the other clips in the stack.



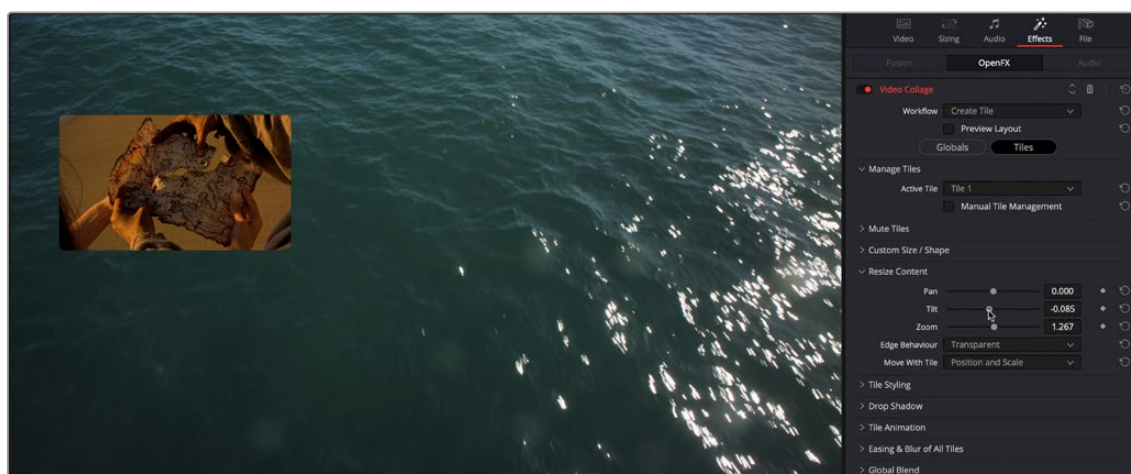
Turning on Preview Layout to customize the layout

With Preview Layout enabled, you can use the Layout controls below to choose how many columns and rows of tiles there are, to stagger the tiles horizontally or vertically, to round or square off each tile by a variable amount, and to alter the margins and spacing. In this way, you can create a wide variety of different symmetrical layouts really easily. In this example, we'll create a layout with one row, three columns, that's staggered vertically.



Our new customized layout, shown next to the Inspector controls for this effect

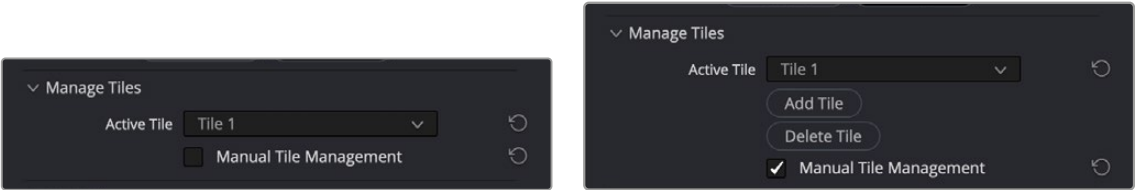
Now that this is done, we'll turn off Preview Layout to focus on the current clip's integration into our layout, and we can see that it doesn't quite fit into the box. Click the Tiles button in the Inspector. This reveals the Tile-specific controls, which include the Resize Content controls. Click the disclosure control to open these up, and use the Pan, Tilt, and Zoom controls to make this image fit into the tile full screen.



Tiles controls in the Inspector with Preview Layout disabled, so we can focus on customizing the current tile

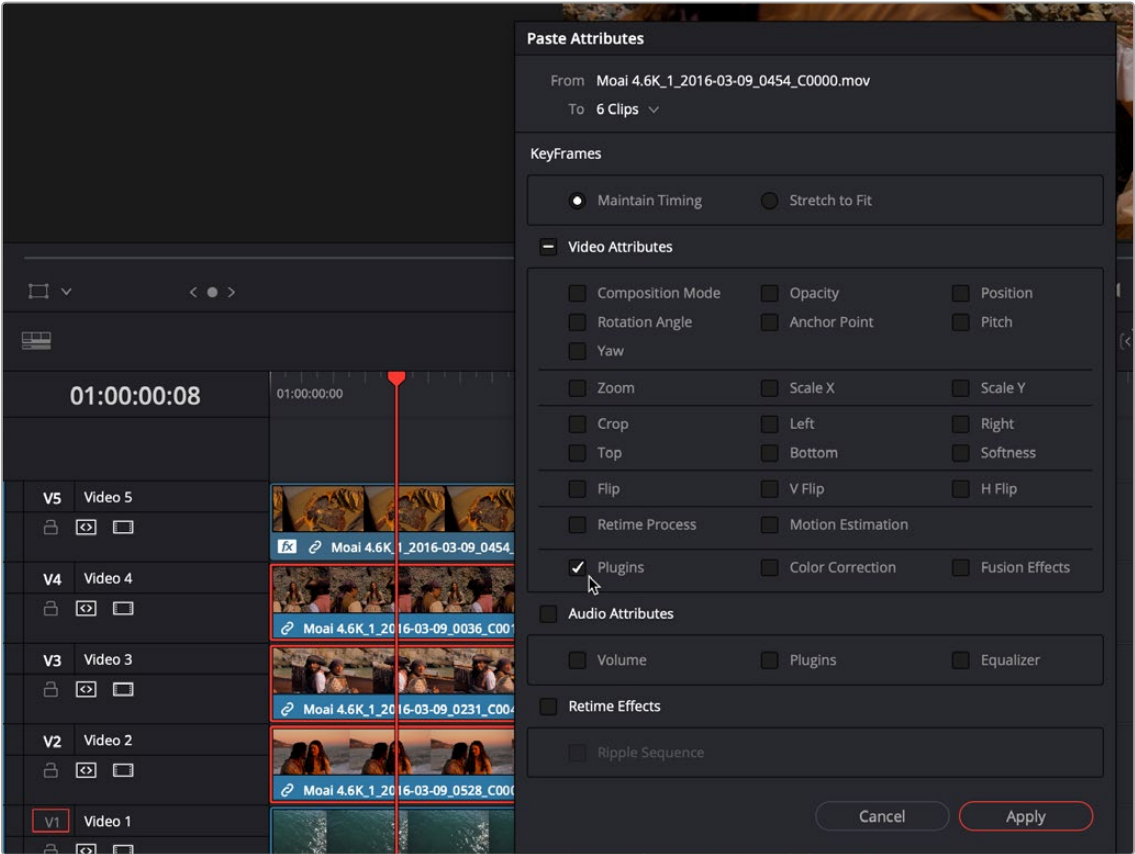
You should also note the Active Tile drop-down menu. This menu serves two purposes. First, it defines which tile the current clip is displayed within. This becomes important later on when you apply this effect to multiple superimposed clips. However, it also determines which tile you further customize using the Tile-based controls of the Inspector. If needed, tiles can be individually styled.

The Manual Tile Management checkbox is off by default, which enables DaVinci Resolve to automatically update the number of tile entries in the Active Tile drop-down menu with items equal to the number of grid positions (or grid nodes) defined by the rows and columns you've set, without you needing to do anything. For simple symmetrical layouts, this is easiest. However, if you turn Manual Tile Management on, you gain the ability to add and delete tile positions manually, which enables other options. For now, we'll leave this turned off.



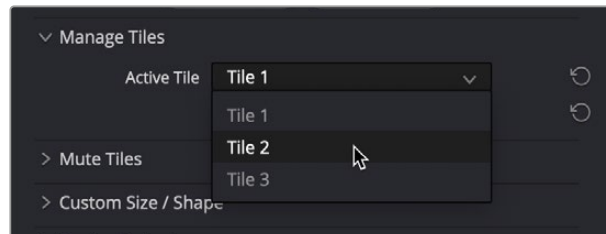
(Left) The Active Tile drop-down with Manual Tile Management turned off, (Right) The Active Tile controls with Manual Tile Management turned on, and additional controls for manual tile management displayed

So, now we've created a custom overall layout, and we've tailored the top clip in the stack to fit the first tile in the layout. Now it's time to apply this effect to the next two clips in the stack (remember the bottom clip will be used as a background, so it doesn't need this plugin applied to it). Select the top clip with the effect applied to it in the Timeline, and press Command-C to copy it. Then, select the next two clips, right-click one of them, and choose Paste Attributes (Option-V). When the Paste Attributes window appears, turn on the Plugins checkbox, in Video Attributes and click Apply. Those two clips disappear as they receive the Video Montage effect and fit themselves into Tile 1, just like the top clip.



Using Paste Attributes to apply the copied effect from the top clip to the other clips in this arrangement

To assign each clip to the correct tile in the layout, select each clip in the arrangement, one at a time, open the Effects panel of the Inspector, click the Tiles button in the Inspector (if necessary), and set each clip to be the next available tile in the Active Tile drop-down menu. In this example, the top clip was in Tile 1, the second clip down will be in Tile 2, and the third clip down will be in Tile 3.



Assigning the Active Tile for each layer in the Timeline

TIP: If you enable the Open FX Overlay of the Timeline Viewer, each tile has an overlay that lets you click a tile in the Viewer to make that the Active Tile.

If you like, you can rearrange which clips go into which tiles using this control, but for now we'll keep things simple. As you do this, each tile becomes populated, and the final layout is revealed, with the background clip showing behind everything. If necessary, you can use the Resize Content controls (in Tiles mode) to resize each tile's contents to fit the tile better.



The current layout, with each clip in the stack assigned to a different tile

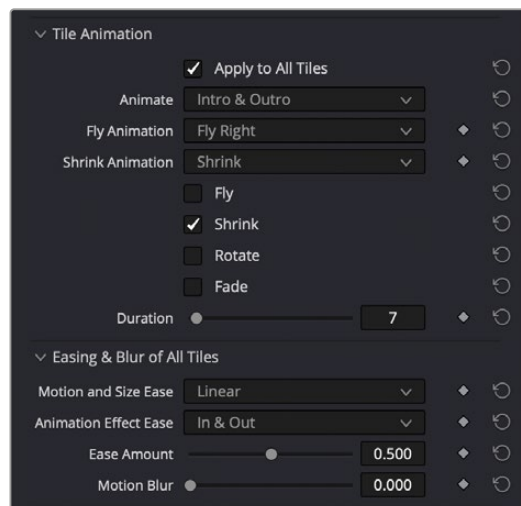
NOTE: If you've customized a layout, copied and pasted attributes to the other clips you want to use in that layout, and have assigned each clip to its position in the layout, and you then decide you need to make a change to the Globals layout controls, you'll need to go through the process of copying the updated effect, pasting attributes to the other clips, and reassigning each clip to the appropriate position in the layout to make that update to the other clips as well.

So, these are the overall workflows of the Video Collage effect. There is, of course, much more you can do, but this is the basis of how you create your initial layout.

Animating Tiles and Advanced Layouts

One of the more powerful aspects of the Video Collage effect is the ability to quickly and easily create animated intros and outros for each tile. These effects can be fully automated over a customizable duration, or their duration can be manually keyframed. All animations can be automatically eased, and you can also enable motion blur for the Fly, Shrink, and Rotate animations.

When you use these controls in Create Background mode, the transparent holes through which other clips in the background show through are animated to “open or close” to reveal the contents. When used in Create Tile mode, the actual tile and its contents are animated to fly, shrink, rotate, and/or fade to appear or disappear.



Tile Animation controls let you create automatic Intro and/or Outro animations.

Additionally, you have the ability to keyframe the Mute, Start and End Column and Row parameters, Tile Styling and Drop Shadow parameters, and other aspects of tiles in Tiles mode, either individually or all together.

In particular, the Start and End Column and Row parameters let you create tiles that span multiple grid positions, and also move tiles from one grid position to another. Without keyframing, this lets you create more kinds of layouts with asymmetric arrangements of large and small tiles. With keyframing, you gain the ability to easily create sophisticated grid-based animations to move tiles from position to position, and to have tiles expand and contract to encompass neighboring tiles in the layout, all with a minimum of keyframing.



Tile Animation controls let you create automatic Intro and/or Outro animations.

Keep in mind that, when using multiple instances of the Video Collage effect on superimposed clips in the Timeline, once you've defined your overall layout and used Paste Attributes to copy a particular instance of this effect with your layout to the rest of the clips, you need only keyframe each instance of each tile within the clip it's applied to. Furthermore, since Intro and Outro animations are applied to the entire duration of the clip, you can easily offset each Intro and Outro animation by offsetting the clips in the Timeline.

TIP: If you change the duration of clips in a stack that are creating a Video Collage effect, you can use the Synchronize Keyframing controls in Globals mode to offset all keyframes applied to that instance of the Video Collage effect by the same amount. You could do the same thing by opening the keyframe track or Curve Editor for all Video Collage parameters in the Timeline and offsetting them that way, but this provides a quick shortcut.

Video Collage Controls

Here's a detailed explanation of all the parameters found in the Video Collage effect.

- **Workflow:** Defines what the output of Video Collage is, depending on how you want to use it.
 - Create Background:** Outputs a frame with holes, behind which you can position individual layers of video to show through.
 - Create Tile:** Outputs a single tile, transformed and styled to be one element of a multi-tile layout
- **Preview Layout:** Turning on Preview Layout lets you see a graphical preview of the entire layout, and how each tile fits together. In Create Tile mode, this lets you configure how you want the layout to look before you copy the clip and use paste attributes to apply this effect to the rest of the video layers you want to arrange into a split-screen.
- **Globals/Tile buttons:** Choose between the Globals controls, which expose controls that let you adjust the overall layout, and the Tiles controls, which let you choose the currently active tile and style it.

Layout

These controls appear in Globals mode.

- **Columns:** Choose how many vertical columns worth of tiles there are.
- **Rows:** Choose how many horizontal rows worth of tiles there are.
- **Stagger Horizontally:** Lets you create an offset from one row to the next in a balanced way, so the rows are all staggered from one another.
- **Stagger Vertically:** Lets you create an offset from one column to the next in a balanced way, so the columns are all staggered from one another.
- **Rounding:** Rounds the edges of each tile in the layout. A value of 0.000 sets the edges to be square. Increasing rounding increasingly curves the corners of each tile, until a value of 1 sets the tile to be perfectly round.

Margins and Spacing

These controls appear in Globals mode.

- **Left/Right Margins:** Alters the spacing from the edge of the frame to the left and right edges of the group of all tiles. Increasing this parameter makes all tiles progressively more narrow, as they squeeze in towards the center of the frame.
- **Top/Bottom Margins:** Alters the spacing from the edge of the frame to the top and bottom edges of the group of all tiles. Increasing this parameter makes all tiles progressively shorter, as they squeeze in towards the center of the frame.
- **Horizontal and Vertical Offset:** These two sliders let you offset the group of all tiles from the center however you like.
- **Horizontal and Vertical Spacing:** These two sliders increase or decrease the spacing between each tile, making them narrower or wider as a result.

Synchronize Keyframing

These controls appear in Globals mode. They're basically a built-in utility that lets you offset all Globals and Tiles parameter keyframes that are applied to the current effect, to account for times when you've done elaborate keyframing to create animated split-screen effects, and you then need to change the duration of the clip this effect is applied to. Instead of forcing you to expose the keyframes of each parameter in the Timeline in order to move them, these controls let you move all keyframes for all parameters forward or backward by a particular number of frames, all at once.

- **Offset Frames:** Lets you specify how many frames you want to offset all keyframes applied to this effect forward. Negative values move keyframes to the left.
- **Apply Keyframing Offset:** Executes the offset.

Manage Tiles

These controls appear in Tiles mode.

- **Active Tile:** This menu serves two purposes. First, it defines which tile the current clip is displayed within. This becomes important later on when you apply this effect to other superimposed clips you want to use together in a layout. However, this control also determines which tile you further customize using the Tile-based controls of the Inspector. If you enable the Open FX Overlay of the Timeline Viewer, each tile has its own overlay that lets you click a tile in the Viewer to make that the Active Tile.
- **Bring Tile Forward/Send Tile Back:** These two buttons let you rearrange the order of tiles in the Active Tile drop-down menu.
- **Manual Tile Management:** While this checkbox is turned off (the default), changing the Columns and Rows parameters automatically updates the Active Tile drop-down menu with one tile per node on the resulting grid arrangement. Turning this checkbox on lets you Add and Delete tiles manually, which lets you create more advanced layouts with overlapping tiles and asymmetric spacing between tiles.

Mute Tiles

These controls appear in Tiles mode. A single checkbox parameter, Mute Tile, lets you enable and disable the visibility of the active tile. This checkbox is keyframable, so you can animate the appearance and disappearance of different tiles to create different effects.

Custom Size/Shape

These controls appear in Tiles mode. They let you further customize layouts on a tile-by-tile basis and can also be used to quickly and easily animate a single tile to move and resize itself around the grid of the overall layout.

- **Start/End Column and Row:** These four controls serve two purposes, letting you change a tile's position by row and column number, or enlarge a tile to encompass multiple tile row or column positions ("spanning"). To use these parameters, you must first choose which tile you want to adjust from the Active Tile drop-down menu. With the tile you want to edit active, you can then drag the appropriate sliders to create either effect, as explained below.

To change Tile Position: When adjusted together so that the Start and End values are identical, they let you manually adjust the position of each tile in the grid that's defined by the number of rows and columns you've set. This is particularly useful when you want to quickly keyframe animated changes to tile position, in order to jump a tile around the grid of the layout.

To make a tile span multiple positions: Setting the Start and End Row and/or Column parameters to different values changes the currently active tile to span multiple grid positions in the layout, making that tile bigger in order to create asymmetric arrangements of larger and smaller tiles. You can also keyframe these values to easily animate the effect of a tile expanding to occupy more grid positions, or shrinking from occupying several grid positions to occupying a single position.

Choose how many rows/columns and in which directions you want that tile to span. Note that if you span multiple rows or columns that have been offset, the resulting tile is enlarged to encompass the entire rectangular region defined by the outer boundaries of both offset tiles.

- **Custom Size/Shape:** Turning this checkbox on enables the rest of the controls in this group, enabling you to set individual scale and rounding for the active tile that's selected.
- **Custom Scale:** Lets you adjust the size of the active tile independently of the other tiles.
- **Custom Rounding:** Lets you adjust the rounding of the active tile independently of the other tiles.

Resize Content

These controls appear in Tiles mode. They let you resize the image being fit within each tile to show exactly what you want shown.

- **Pan/Tilt/Zoom:** These three sliders let you move and resize the video layer shown inside of the active tile to fit however you need it to.
- **Edge Behavior:** These settings allow you to control how the edges of the video layer respond inside the tile.

Transparent: Any positioning that goes past the edge of the video is transparent, showing the Tile Color set in the Tile Styling options.

Reflect: Any positioning that goes past the edge of the video mirrors back on itself.

Wrap: Any positioning that goes past the edge of the video repeats itself.

Replicate: Any positioning that goes past the edge of the video shows a duplicate of the edge stretched to the boundary of the control, essentially filling in the remaining space with the pixels used at the extreme edges of the video clip.

- **Move With Tile:** Defines how the image is fit into the tile when the tile size changes due to other parameters of the Video Collage effect.

Position Only: The video content in the tile only is sized using Pan and Tilt; Zoom is ignored.

Position and Scale: The video content in the tile is sized using Pan, Tilt, and Zoom.

Tile Styling

These controls appear in Tiles mode. They let you add a colored border and adjust the opacity of all tiles or just the currently active tile. These parameters are animatable.

- **Apply to All Tiles:** When checked, this applies any changes made in the Tile Styling parameters to all tiles in the Video Collage, regardless of which tile is currently active.
- **Tile Border:** This slider controls the width of the tile's border. It has a possible range of values from -0.250 to 0.250, where negative numbers expand the size of the border inside the tile, and positive numbers expand it outside the tile.
- **Tile Color:** Allows you to choose the background and border color of the tile.
- **Tile Opacity:** This slider changes the transparency of the tile itself; it does not effect the video clip inside the tile. The possible ranges are 0.000 (fully transparent) to 1.000 (fully opaque).

Drop Shadow

These controls appear in Tiles mode. They let you add and customize a drop shadow, either to all tiles or just to the currently active tile. These parameters are animatable.

- **Apply to All Tiles:** When checked, this applies any changes made in the Drop Shadow parameters to all tiles in the Video Collage, regardless of which tile is currently active.
- **Strength:** This slider controls drop shadow transparency. The possible ranges are 0.000 (fully transparent) to 1.000 (fully opaque).
- **Color:** Allows you to choose the color of the Drop Shadow.
- **Drop Angle:** Lets you set the angle that the shadow is cast from. The slider represents 360 degrees of movement from -180.0 to 180.0.
- **Drop Distance:** Allows you to set the distance that the shadow appears from its source tile. The possible values are from 0.000 to 0.200.
- **Expand:** This slider controls how far the shadow expands from its origin. The possible values are from 0.900 to 1.250.
- **Blur:** Allows you to control the amount of diffusion of the Drop Shadow. The possible ranges are 0.000 (no blur) to 1.000.

Tile Animation

These controls appear in Tiles mode. They make it easy to create simple intro/outro animations to bring all or some tiles onscreen, and to push all or some tiles offscreen, in different ways. The Animate drop-down menu determines whether this animation will be automatic or manually created using keyframes.

- **Apply to All Tiles:** When enabled, any animation that's set up will be applied to all tiles in the layout simultaneously.
- **Animate:** This drop-down menu lets you choose between Manually Keyframing intro or outro animations or automatically creating animations to introduce or conclude a tile. The options are:

Manually Keyframe: In this mode, you must manually keyframe the four progress sliders below to create animated Intro and Outro animations.

Intro Only: Animation is automatically created to bring each tile onscreen according to the types of animation you choose from four checkboxes below. Options include: Fly, Shrink, Rotate, and Fade, in any combination. A Duration slider appears that lets you choose how long the Intro will be.

Outro Only: Animation is automatically created to send each tile offscreen according to the types of animation you choose from four checkboxes below. Options include: Fly, Shrink, Rotate, and Fade, in any combination. A Duration slider appears that lets you choose how long the Outro will be.

Intro & Outro: Animation is automatically created to bring each tile onscreen and then send each tile offscreen, according to the types of animation you choose from four checkboxes below. Options include: Fly, Shrink, Rotate, and Fade, in any combination. A Duration slider appears that lets you choose how long the Intro and Outro will be.

Fly Animation: Lets you choose a direction for tiles to fly in from and fly out to.

Shrink Animation: Lets you choose a method of shrinking out or in when the Fly Progress slider is animated.

Fly Progress: (manual key framing only) A slider that, when animated, lets you fly a tile into or out of the frame. A value of 0.000 positions the tile in its default grid position. A value of 1.000 positions the tile entirely off screen.

Shrink Progress: (manual key framing only) A slider that, when animated, lets you shrink a tile down to nothing or up to the current size. A value of 0.000 sizes the tile at its largest state. A value of 1.000 shrinks the tile to nothing.

Rotate Progress: (manual key framing only) A slider that, when animated, lets you fly a tile into or out of the frame. A value of 0.000 positions the tile in its default grid position. A value of 1.000 positions the tile entirely off screen.

Fade Progress: (manual key framing only) A slider that, when animated, lets you fly a tile into or out of the frame. A value of 0.000 positions the tile in its default grid position. A value of 1.000 positions the tile entirely off screen.

Easing & Blur

These controls appear in Tiles mode. They let you adjust the easing of the tile animation created in the previous group of controls.

- **Motion and Size Ease:** These controls set the curve used for simulated momentum in animating the Tile Position and Size keyframes of the tiles.
 - None:** No curves are applied. The animation snaps immediately from keyframe to keyframe.
 - Linear:** No easing curves are applied; motion is neither accelerated or decelerated.
 - In:** The animation curve starts slower and ends faster.
 - Out:** The animation curve starts faster and ends slower.
 - In & Out:** The animation curve starts slower, accelerates in the middle, and slows down at the end.
- **Animation Effects Ease:** These controls set the curve used for simulated momentum in animating the Tile Animation keyframes of the tiles.
 - None:** No curves are applied. The animation snaps immediately from keyframe to keyframe.
 - In:** The animation curve starts slower and ends faster.
 - Out:** The animation curve starts faster and ends slower.
 - In & Out:** The animation curve starts slower, accelerates in the middle, and slows down again at the end.
- **Ease Amount:** The relative strength of the animation curve in speeding up and slowing down. The values range from 0.000 to 1.000.
- **Motion Blur:** This slider controls the amount of motion blur added to the animation, in order to make it smoother and more organic looking. The values range from 0.000 (no motion blur), to 1.000 (maximum motion blur).

Global Blend

- **Blend:** This slider controls the overall opacity of the entire Video Collage effect. The values range from 0.000 (completely transparent) to 1.000 (completely opaque).